

# DIRECTING SILICON ENGINEERS

*What Product Management Still Buys You When the Engineer Is Made of Silicon*

## ACT ONE

# WHAT YOU ALREADY DO — AND WHAT'S ACTUALLY DIFFERENT

0:40 – 1:10 (8 min)

*Setting up the distinction before Act Two does anything with it.*

# Before We Start

Three personal projects. I'm the only customer on all of them.

Chat-based AI assistants and agentic tools, side by side.

My own mistakes are in this deck, on purpose.

*The claim: these patterns hold whether the customer is one guy with a golf habit or a Fortune 15 delivery org.  
The rest of this talk is the evidence for that claim.*

# Two Ways You're Probably Already Working With AI

## CHAT-BASED AI

A back-and-forth conversation — ChatGPT, Claude.ai, Copilot Chat. You ask, it answers, you decide what to do next. Most of your AI experience today: sharpening AC, drafting Gherkin, checking a document.

## AGENTIC AI

You describe a goal once — Claude Code, Codex, Antigravity — and it takes many steps on its own before coming back to you. This talk calls that a “silicon engineer.”

# How the Work Happens

## BEFORE

You ask. It answers. You decide.

## AFTER

You write the rules once. It acts for hours without you watching.

# What a Bad Call Costs You

## BEFORE

A bad suggestion costs nothing — you just don't use it.

## AFTER

A bad call might already be saved into the project and acted on before you ever see it.

# Where the Memory Lives

## BEFORE

The conversation is the memory.

## AFTER

The conversation ends. The agent forgets everything. The memory has to live in the project's own files, not the chat.

# Who Else Is in the Room

## BEFORE

One model, one thread.

## AFTER

Claude Code, Codex, and Antigravity may all touch the same project on different days, with no shared memory between them. Whatever you wrote down is the only thing they have in common.



# What a Standing Contract Actually Looks Like

*Six lines from a real AGENTS.md – the file an agent reads every session, not a prompt you type once. (“Coverage” below just means how much of the code the automated tests actually check.)*

- \* 100% Core Coverage Target: Maintain test coverage targeting 100% (hard floor of 95%) for all core state engines.
- \* Zero-Defects & Warnings: Handover is blocked if there are compile errors, TypeScript warnings, or failing tests.
- \* Human-in-the-Loop Intervention: If an implementation requires falling below the 95% coverage threshold or leaving warnings unresolved, the coding agent **MUST** pause and request explicit approval from the Product Owner.

## ACT TWO

# WHAT YOUR PM/PO SKILLS BECOME WHEN YOU'RE DIRECTING AN AGENT

8:00 – 8:30 (13 min)

*Some of this you already have. Some of it is genuinely new muscle.*

# Already in Your Toolkit

- Turning intent into a clear, checkable spec — the same skill as sharpening AC in chat, just aimed at a document the agent reads every time it starts work.
- Insisting on real evidence, not just confidence — the same instinct as pushing back when someone says “trust me, it's fine” instead of showing you why.
- Saying what's out of bounds — the same “here's what we're not doing” instinct you already have, with a new reason it matters: the AI doesn't feel the cost of doing too much, the way a human developer would.

## A Definition of Done the Agent Can't Talk Its Way Past

*Not a checklist you eyeball. A gate it literally cannot pass.*

- Kanban Game: work isn't “done” unless automated tests check at least 95% of the code — a hard “coverage floor”
- Job Search: the automated packaging step itself refuses to finish if any test fails
- Golf OS: an automatic check that refuses to accept a document if a required section is missing

# Decide What the Agent Never Decides Alone

## THE METRIC

Kanban Game: if tests stop checking at least 95% of the code, stop and ask. A hard number, checked automatically.

## THE HEURISTIC

Golf OS: “Would Braz want to have had a say in this?” A judgment call, applied to any decision point that isn't routine.

*Two valid ways to encode the same guardrail – the point is naming the trigger before the agent acts, not after.*

## Design for the Agent's Amnesia

*The session ends. The agent forgets everything. And you may not be the only tool touching this project.*

- TASK.md checklists and WORKLOG.md handoff notes (Kanban Game, Job Search)
- Golf OS keeps “the official estimate on record” and “the agent's own guess” separate on purpose — neither silently overwrites the other
- A reversed recommendation gets marked “superseded,” not deleted — so a later session can see the answer changed, and why

# Job Size Just Grew a Second Unit of Measure

WSJF prices a Story by Cost of Delay over Job Size. Job Size has always meant one thing: relative effort.

*Agentic tools hand you a second, metered cost that doesn't always move with effort – a small ask can burn a large share of quota, and a big ask can be cheap if it's mechanical.*

## Anti-Pattern: Scope-Bounding, Skipped

A rule I wrote for one project got copied into a second, mirrored copy of that project — where the opposite policy applied. A change landed in the main version without review before I caught it.

*The fix wasn't “pay closer attention.” It was making the rule name its own scope, in its own text.*



## Anti-Pattern: Evidence Discipline, Skipped

Golf OS forces every recommendation to state how confident it is. My Job Search project's agent handed me a “95/100 match score” against undisclosed comp — zero caveat.

*Same person, same standards I actually hold. The agent only had the discipline I'd written down.*

*The discipline doesn't travel by osmosis.  
It only holds where you actually wrote it down –  
and the agent can see it.  
Including from yourself, project to project.*

## ACT THREE

# WHAT TO GO DO ABOUT IT

21:00 – 21:30 (9 min)

*Self-investment, and why it's a career move, not a hobby.*

# Reframe the Threat

The part of delivery that used to require a human developer to execute a spec is exactly what agentic AI is fastest at automating.

*The scarce role is the person who writes the contract, sets the guardrails, and verifies the outcome. That's a PM/PO job description – not a developer one.*

# What It Actually Costs to Build This Yourself

| Tool                    | Entry          | Mid               | Top                     |
|-------------------------|----------------|-------------------|-------------------------|
| Claude Code (Anthropic) | Pro — \$20/mo  | Max 5x — \$100/mo | Max 20x — \$200/mo      |
| Codex (OpenAI)          | Plus — \$20/mo | Pro 5x — \$100/mo | Pro 20x — \$200/mo      |
| Antigravity (Google)    | Free — \$0/mo  | AI Pro — ~\$20/mo | AI Ultra — \$100-200/mo |

Prices as of Aug 2026. Consider this the “market price” line on a restaurant menu — check today's before you order.

# The Actual Gap, If You're Chat-Only

- You already have the spec-writing and evidence skills from Act Two.
- What's missing is hands-on time with a few specific mechanics.
- Reading a “diff” — the side-by-side view of exactly what changed in a file.
- Understanding a “build gate” — an automatic checkpoint that blocks progress if a rule isn't met.
- Enough comfort with the tools that track changes to review a proposed change without help.
- Recognizing when confident-sounding output is actually thin.

# This Week

- 1 Rewrite one AC as a standing rule an agent would have to satisfy before calling the work done.
- 2 Run one low-stakes agentic session end-to-end. Notice the moment you'd have wanted a stop-and-ask gate.
- 3 Learn to read a diff and a pass/fail test result well enough to tell if “done” means done.
- 4 Draft a one-page working agreement for your own team's next AI-assisted effort.
- 5 Start one side project to build the muscle — the same \$20/month it costs to find out if you like it.

*“The work-print cut of this movie used Impact for its titles – fast, obvious, technically fine. The theatrical cut used one typeface, deployed with total attention to every small detail, and it's the version that still gives you chills forty-plus years later. Agentic AI will hand you the Impact version of anything in about four seconds. The Goudy Oldstyle version – the one that's actually amazing – still needs someone paying attention to the small stuff.”*

– Michael Lopp, “[The Title Cards in Blade Runner Are Fucking Amazing](#),” Aug 2026



# QUESTIONS & ANSWERS

29:30 – 30:00

*The live talk ends here – everything after this is reference.*



APPENDIX

# REFERENCE MATERIAL AND BACKUP

*Not part of the live talk – held for questions and as a leave-behind.*



## Habits I Didn't Plan to Repeat (1/2)

- Write the rulebook, not just the request – acceptance criteria stop being a conversation starter and become the literal spec the agent checks its own work against before it's allowed to call something done.
- Make “done” something the system checks, not something someone eyeballs – a Definition of Done becomes a gate that blocks the work from finishing if it isn't met, not a checklist that's easy to wave through under deadline pressure.
- Say what's out of bounds, in writing, every time – all three projects reinvented the same habit independently: write down the limit – a budget, a rate cap, a flat “we're not doing this” – so the agent isn't left guessing where the edges are.
- Decide in advance what needs a human's sign-off – sometimes a hard number that trips a stop, sometimes a judgment call written as a question. Either way, the fork gets named before the agent reaches it, not after.

## Habits I Didn't Plan to Repeat (2/2)

- Leave a note for whoever picks this up next – including yourself – a running log of what happened and why, so a different session, a different day, or even a different AI tool can continue the work without you re-explaining it from scratch.
- Make every claim say how sure it is – recommendations get labeled by confidence, so a guess and a well-supported finding never read the same way on the page.
- Cost isn't just effort anymore – it's also usage – alongside “how much work is this,” there's now “how much of my metered AI usage will this burn” – and those two numbers don't always agree.
- The discipline only holds where you actually wrote it down – when a rule existed in one project but not its near-identical twin, the gap showed up immediately. Same author, same standards – different results, because only one contract had the words on the page.

# Chat-LLM Assist vs. Agentic AI

|                 | Chat-LLM Assist        | Agentic AI                                 |
|-----------------|------------------------|--------------------------------------------|
| Unit of work    | One turn: ask, answer  | A working loop: claim, build, verify, ship |
| Who acts        | You, after reading it  | The agent – files, commits, sometimes PRs  |
| Memory          | None beyond this chat  | Must be externalized – the session forgets |
| Governed by     | Your prompt, right now | A standing document read every session     |
| Bad call caught | Immediately, by you    | Possibly after it already acted            |

## Turn Budget and Estimate Provenance

- Kanban Game Turn-Budget table:
  - XS/S: 1-3 turns, ~ 1-10% of weekly quota
  - M/L: 4-8 turns, ~ 5-25% of weekly quota
  - XL: 8-12 turns, ~ 30-50% of weekly quota
  - XXL: 12+ turns – blocked from starting until split
- Golf OS Estimate Reporting order:
  - 1. Repo estimate – the durable point estimate
  - 2. Agent estimate – this session's own guess
  - 3. Weekly quota % – that guess as a share of quota
  - 4. Quota basis – the explicit assumption behind it
  - Neither number is allowed to silently replace the other.

## ACT ONE – THE ACTUAL DOCUMENT

# AGENTS.md, in Full — Kanban Game

*The complete policy set the live teaser only sampled from.*

### ## Agentic Test-Driven Development (TDD) Policy

- \* BDD-First & Test-First Implementation: all development work must start with Gherkin scenarios and step definition tests before features or core changes.
- \* Business Stakeholder Translation: treat the Product Owner as a business stakeholder; translate conceptual goals into testable Given-When-Then scenarios.
- \* Verify Red Phase: confirm tests fail before writing code to make them pass.

### ## Free-Tier & Platform Design Constraints

- \* Spark Tier Conservation: keep reads/writes minimal to fit the Firebase Spark free tier (50k reads / 20k writes per day). Avoid polling; clean up listeners.
- \* No background servers: the app runs fully serverless.

## Glossary: The Basics

- Repo (repository) — *the actual folder of a project's files and history; where an agent's instructions and its work both live.*
- AGENTS.md / CLAUDE.md — *the file an agentic tool reads at the start of every session, spelling out its rules.*
- Commit — *a saved snapshot of one change to a project.*
- Merge — *folding a finished change into the main version of a project.*
- Pull request (PR) — *a proposed change, held for review before it's merged.*
- Diff — *a side-by-side view of exactly what changed in a file.*
- Build gate — *an automatic checkpoint that blocks progress unless a rule (like “tests must pass”) is met.*
- Test coverage — *how much of the code automated tests actually check, usually shown as a percentage.*



## Glossary: This Talk's Own Terms

- Working loop — *claim a piece of work, build it, verify it, ship it, close it.*
- Handoff note — *a short written record letting a different session resume with no shared memory.*
- Resync — *re-reading the working agreement after it's changed while you were away.*
- Turn — *one step an agent takes; a story's “turn budget” is how many steps it's expected to cost.*
- Token — *the unit an AI model's usage is metered in; loosely, a fragment of text it reads or writes.*
- WSJF (Weighted Shortest Job First) — *a prioritization formula:  $\text{Cost of Delay} \div \text{Job Size}$ .*
- Decision Surfacing — *pausing on a judgment call and explaining it before acting, not after.*
- Silicon engineer(s) — *this talk's term for an agentic coding tool.*

## This Week (Handout Version)

- 1 Rewrite one AC as a standing rule an agent would have to satisfy.
- 2 Run one low-stakes agentic session end-to-end; notice the stop-and-ask moment.
- 3 Learn to read a git diff and a pass/fail test result.
- 4 Draft a one-page working agreement for your team's next AI-assisted effort.
- 5 Start one side project — the same \$20/month it costs to find out if you like it.

## Anti-Pattern Citations

- A safety rule I wrote for one project – “merge automatically once tests pass” – got copied into a near-identical second copy of that project, where the opposite rule was supposed to apply. Because the rule never said which project it belonged to, a change landed in the live version without the review it should have gotten. (*Job Search project, work log entry, July 2026.*)
- One of my own projects handed me a job-match score of “95 out of 100” with no mention that the salary was unknown – a confident-sounding number with zero caveat. A different project of mine requires every recommendation to state its own confidence level. Same person, same standards I actually hold – the discipline only existed where I'd written it into the rules. (*Job Search project's match report, versus Golf OS's confidence-field requirement.*)